NITTE (Deemed to be University) | NMAM INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

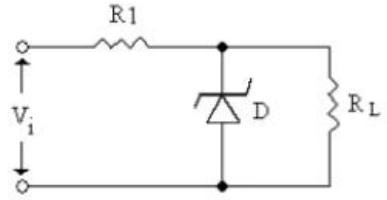
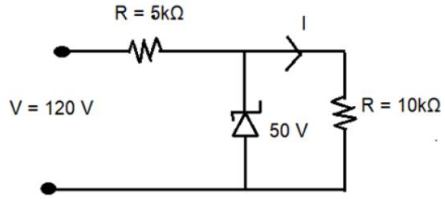
COURSE: BASIC ELECTRONICS (MCQ)

COURSE CODE: ECE111

UNIT -I

1	The voltage at which forward current through the diode starts increasing rapidly is called as
a	Cut in voltage
b	Breakdown voltage
c	Saturation voltage
d	Cut off voltage
Ans	a
2	Smaller the ripple factor, the output will have higher components of
a	Spike
b	AC
c	DC
d	pulse
Ans	c
3	The efficiency of full wave rectifier is about
a	0.46
b	81.2
c	1.21
d	40.6
Ans	b
4	When the p-n junction diode is forward biased, it offers
a	High resistance
b	Low resistance
c	Low voltage
d	High voltage
Ans	b
5	Zener diode is mainly used as
a	Amplifier
b	Comparator
c	Oscillator
d	Voltage regulator
Ans	d
6	The efficiency of half wave rectifier is about
a	81.2%
b	46 %
c	40.6 %

d	1.21%
Ans	c
7	The r.m.s value of load current in a full wave rectifier is
a	0.707 I_m
b	0.5π
c	0.5 I_m
d	0.3183 I_m
Ans	a
8	The amount of ripples present at the output of half wave rectifier is
a	1.21
b	0.81
c	0.46
d	0.50
Ans	a
9	The amount of ripples present at the output of a full wave rectifier is
a	81%
b	48%
c	42%
d	52%
Ans	b
10	A silicon diode is connected in series to a dc supply of 2 volts and resistance of 13 Ω, the current in the circuit is
a	1.5A
b	0.55 A
c	0.45 A
d	0.10A
Ans	d
11	The DC load current of a half wave rectifier is
a	2Idc/π
b	2Im/π
c	Im/π
d	1.414/π
Ans	c
12	A sinusoidal voltage of peak value 40V is applied to a half wave rectifier with R_L = 800 Ω and R_f = 8 Ω. The peak value of the current is
a	4.95 mA
b	49.5 mA
c	0.495 mA
d	4.95 A
Ans	b
13	A silicon diode is connected in series to a dc supply of 5 Volts and resistance of 1.5 kΩ, the current in the circuit is
a	3.86 mA

b	2.86 mA
c	0.38 mA
d	1.5mA
Ans	b
14	Average DC load voltage of a full wave Bridge rectifier is
a	Im/0.5
b	2Im/π
c	Im/0.707
d	2Vm/π
Ans	d
15	In the voltage regulator shown below, if the current through the load decreases 
a	The current through R1 will increase.
b	The current through R1 will decrease.
c	Zener diode current will increase.
d	Zener diode current will decrease
Ans	c
16	In the circuit, what is the output voltage? 
a	50 V
b	70 V
c	120 V
d	170 V
Ans	a
17	The r.m.s value of load current in a half wave rectifier is
a	0.5 I_m
b	0.5π
c	0.707 I_m
d	0.3183 I_m
Ans	a
18	A germanium diode is connected in series to a dc supply of 4volts and resistance of 2 kΩ, the current in the circuit is
a	1.85 mA
b	1.45mA
c	0.185 mA

d	18.5 mA
Ans	a
19	A zener diode of 10V is connected to a supply of 20V through a series resistance of 200 ohms. With no load, current through the zener is
a	500 mA
b	50 mA
c	5 A
d	400 mA
Ans	b
20	A germanium diode is connected in series to a dc supply of 7.5 volts and resistance of 1.3 kΩ, the current in the circuit is
a	5.53 mA
b	5.53 A
c	5.35 mA
d	1.53 mA
Ans	a
21	A silicon and a germanium diodes are connected in series to a dc supply of 8 volts and resistance of 2.5 kΩ, the current in the circuit is
a	2.8 A
b	28 mA
c	0.28 mA
d	2.8 mA
Ans	d
22	In the breakdown region, zener diode behaves like a source of
a	Constant resistance
b	Constant current
c	Constant voltage
d	Constant power
Ans	c
23	Ripple factor for a full wave bridge rectifier with capacitor filter is
a	$\frac{1}{4\sqrt{3}fCR_L}$
b	$\frac{1}{2\sqrt{3}fCR_L}$
c	$\frac{1}{2fCR_L}$
d	$\frac{1}{4fCR_L}$
Ans	a
24	Which of the following statement is correct?
a	Rectification efficiency of a half wave rectifier is equal to that of the Full wave rectifier
b	All of the given options
c	Rectification efficiency of a half wave rectifier is lower than that of the Full wave rectifier
d	Rectification efficiency of a half wave rectifier is higher than that of the Full wave rectifier
Ans	c

25	In a Full wave bridge rectifier, the number of diodes conducting at a time during one half cycle of the input signal is
a	2
b	4
c	3
d	1
Ans	a
26	The doping level in a zener diode is …… that of a PN diode.
a	The same as
b	Less than
c	More than
d	None of the given options
Ans	c
27	The basic purpose of filter is to
a	Minimize variations in ac input signal
b	Suppress harmonics in rectified output
c	Remove ripples from the rectified output
d	Stabilize dc output voltage
Ans	c
28	The basic reason why a full wave rectifier has a twice the efficiency of a half wave rectifier is that
a	It makes use of transformer
b	Its ripple factor is much less
c	It utilizes both half-cycle of the input
d	Its output frequency is double the line frequency
Ans	c
29	In a full wave rectifier, if the input frequency is 50 Hz, then output frequency will be
a	50 Hz
b	75 Hz
c	100 Hz
d	200 Hz
Ans	c
30	To get a peak load voltage of 40V out of a bridge rectifier. What is the approximate rms value of secondary voltage?
a	0 V
b	14.4 V
c	28.3 V
d	56.6 V
Ans	c

UNIT -II

1	Total emitter current in BJT is
a	$I_C + I_{CBO}$
b	$I_C + I_E$
c	$I_B + I_C$
d	$I_B – I_C$
Ans	c
2	Which is not a MOSFET terminal?
a	Gate
b	Source
c	Drain
d	Base
Ans	d
3	What is the collector current for a C-E configuration with a beta of 100 and a base current of 30 μA?
a	3μA
b	3 mA
c	30 μA
d	30 mA
Ans	b
4	In a CE amplifier circuit, the output signal generated with a phase shift of
a	180^0
b	90^0
c	0^0
d	270^0
Ans	a
5	For a BJT, the following statement is true
a	$I_B \cong I_E$
b	$I_B \cong I_C$
c	$I_E \cong I_C$
d	$I_B \cong \alpha I_E$
Ans	c
6	Which transistor element is used in CMOS logic.
a	MOSFET
b	JFET
c	BJT
d	Diode
Ans	a
7	A transistor has a beta of 200 and a base current of 40 μA in CE configuration. Then the collector current is
a	8 mA

b	8μA
c	80 μA
d	80 mA
Ans	a
8	BJT is made to work as an amplifier when it is operated in ..... region
a	Inverted mode
b	Saturation
c	Cut off
d	Active
Ans	d
9	A transistor has I_B = 0.08 mA and I_E = 9.60 mA. The value of β is
a	129
b	119
c	960
d	100
Ans	b
10	The enhancement type MOSFET works only with _____.
a	large negative gate voltage
b	large positive source voltage
c	large positive gate voltage
d	large negative drain voltage
Ans	c
11	A transistor has I_B = 0.06 mA and I_E = 9.00 mA. The value of β is
a	149
b	129
c	960
d	100
Ans	a
12	In CE configuration of BJT, output characteristics is plotted by taking
a	V_{CE} verses I_C for constant I_E
b	V_{CE} verses I_C for constant I_B
c	V_{CE} verses I_C for constant V_{CB}
d	V_{CE} verse I_E for constant I_C
Ans	b
13	A transistor has I_B = 0.06 mA and I_E = 9.00 mA. The value of α is
a	9.9
b	0.99
c	990
d	100
Ans	b
14	The MOSFET stands for
a	Metal oxide semiconductor FET
b	Metal oxide surface FET

c	Metal oxidized selenium FET
d	Metal of surface FET
Ans	a
15	When the transistor operates as an amplifier, it is operating at
a	In the active region
b	In the cut-off region
c	In the Ohmic region
d	In the Saturation region
Ans	a
16	When BJT operates as a switch, it is operated in _____
a	Cut-off and Saturation region
b	Active and saturation region
c	Active and cut-off region
d	Saturation and active region
Ans	a
17	The collector to base current gain β in terms of α is defined as
a	$\frac{\alpha}{1-\alpha}$
b	$\frac{\alpha}{1+\alpha}$
c	$\frac{\beta}{1+\alpha}$
d	$\frac{\beta}{1+\beta}$
Ans	a
18	The collector to emitter current gain α in terms of β is defined as
A	$\frac{\beta}{1+\beta}$
B	$\frac{\alpha}{1-\alpha}$
C	$\frac{\beta}{1+\alpha}$
D	$\frac{\alpha}{1+\alpha}$
Ans	a
19	The leakage current in a transistor is due to:
A	Zener effect
B	Majority charge carriers
C	Minority charge carriers
D	Breakdown
Ans	c
20	A bipolar junction transistor is a
A	Current controlled device
B	Voltage controlled device

C	Resistance controlled device
D	Junction controlled device
Ans	b
21	The value of α of a transistor is …….
A	more than 1
B	Less than 1
C	0.76 to 0.96
D	1
Ans	b
22	The value of β for a transistor is generally ……..
A	1
B	Less than 1
C	Between 20 and 500
D	Above 500
Ans	c
23	The value of β for a silicon transistor given the collector current as $1\ mA$ and base current as $10\ \mu A$ is
a	100
b	10
c	0.1
d	1000
Ans	a
24	If an emitter current is changed by $4mA$, the collector current changes by $3.5mA$. The value of β will be:
a	8
b	0.5
c	3.5
d	7
Ans	d
25	A MOSFET is a
a	Current controlled device
b	Voltage controlled device
c	Resistance controlled device
d	Junction controlled device
Ans	b
26	A MOSFET has…….. terminals.
a	Two
b	Five
c	Three
d	Four
Ans	c
27	If the value of α is 0.9, then value of β is …….

a	9
b	0.9
c	900
d	90
Ans	a
28	The value of β for a silicon transistor given the collector current as 1 mA and base current as 25 μA is
a	40
b	4
c	400
d	4000
Ans	a
29	The value of α for a silicon transistor, given the collector current as 1 mA and base current as 25 μA is
a	0.9176
b	0.9756
c	0.9276
d	0.9376
Ans	b
30	In CE-RC coupled amplifier, an emitter bypass capacitor is connected in parallel with emitter resistance R_E to
a	Increase the noise in the circuit
b	To provide a low reactance path to the amplified ac signal
c	To support the resistance R_E
d	To block the dc voltage
Ans	b
31	The number of depletion layers in a transistor is ……
a	Four
b	Three
c	One
d	Two
Ans	d
32	In a transistor, signal is transferred from a …… region
a	high resistance to low resistance
b	low resistance to high resistance
c	high resistance to high resistance
d	low resistance to low resistance
Ans	b
33	With the E-MOSFET, when gate input voltage is zero, drain current is …..
a	At saturation
b	Zero
c	I_{DSS}
d	Widening the channel
Ans	b

34	MOSFET has a thin layer of silicon dioxide, which act as:
a	Resistance
b	Inductor
c	Semiconductor
d	Capacitor
Ans	d
35	The base of a transistor is …….. doped
a	Heavily
b	Moderately
c	Lightly
d	Not
Ans	c
36	The emitter of a transistor is …… doped
A	Heavily
B	Moderately
C	Lightly
D	Not
Ans	a

UNIT -III

1	Which of the following electrical characteristics is not exhibited by an ideal op-amp?
a	Infinite output resistance
b	Infinite bandwidth
c	Infinite voltage gain
d	Infinite slew rate
Ans	a
2	An integrator circuit using an Op Amp has ……..in its feedback path
a	Resistor
b	inductor
c	Capacitor
d	Diode
Ans	c
3	Find the output voltage of an ideal op-amp if V_1 and V_2 are the two input voltages
a	$V_0 = V_1 - V_2$
b	$V_0 = A(V_1 - V_2)$
c	$V_0 = A(V_1 + V_2)$
d	$V_0 = V_1 \times V_2$
Ans	b
4	A differentiator circuit using an Op Amp has ……..in its feedback path

a	Capacitor
b	Resistor
c	Diode
d	Inductor
Ans	b
5	A sine wave of 0.5 V peak voltage is applied as an input to an inverting amplifier with R_1 = 10 kΩ and R_f = 50 kΩ The output voltage V_o is
a	5.2 V peak
b	-10 V peak
c	-2.5 V peak-peak
d	-2.5 V peak
Ans	d
6	The output voltage obtained for an ideal op-amp is by
a	Amplifying individual input voltages
b	Amplifying products of two input voltage
c	Amplifying the difference between the two input voltages
d	None of the mentioned
Ans	c
7	The output voltage V_0 of an Op Amp integrator is given by
a	$V_0 = -\frac{1}{R_1 C_f}\int_0^t V_{in}\,dt$
b	$V_0 = -\frac{1}{C_f}\int_0^t V_{in}\,dt$
c	$V_0 = -\frac{1}{R_1 C_f}\int_{-\infty}^t V_{in}\,dt$
d	$V_0 = -\frac{1}{R_f C_1}\int_0^t V_{in}\,dt$
Ans	a
8	A sine wave of 0.5 V peak voltage is applied as an input to a non- inverting amplifier with R_1 = 10 kΩ and R_f = 50 kΩ The output voltage Vo is
a	3 V peak
b	2.5 V peak
c	-3 V peak
d	3 V peak-peak
Ans	a
9	Which is not the ideal characteristic of an op-amp?
a	Output impedance is zero
b	Input Resistance is zero
c	Bandwidth is infinity
d	Open loop voltage gain is infinity
Ans	b
10	The output voltage V_0 of an Op Amp differentiator is given by

a	$V_0 = -R_f C_1 \frac{dV_{in}}{dt}$
b	$V_0 = -\frac{1}{C_f}\int_0^t V_{in}\,dt$
c	$V_0 = -\frac{1}{R_f}\int_0^t V_{in}\,dt$
d	$V_0 = -\frac{1}{R_f C_1}\frac{dV_{in}}{dt}$
Ans	a
11	A sine wave of 3.5 V peak voltage is applied as an input to a non- inverting amplifier with R_1 = 20 kΩ and R_f = 80 kΩ The output voltage Vo is
a	-12.5 V peak
b	10 V peak
c	17.5 V peak
d	-5 V peak
Ans	c
12	Which factor determines the output voltage of an op-amp?
a	Supply voltage
b	Both positive and negative saturation voltage
c	Negative saturation voltage
d	Positive saturation voltage
Ans	b
13	The output voltage swing of a comparator for an applied input voltage depends on
a	Regulated power supply voltages
b	Dual power supply voltages
c	AC signals applied at the terminals of the Op Amp
d	DC signals applied at the terminals of the Op Amp
Ans	b
14	A sine wave of 2 V peak voltage is applied as an input to an inverting amplifier with R_1 = 24 kΩ and R_f = 82 kΩ The output voltage Vo is
a	-8.833 V peak
b	-8.833 V peak-peak
c	-6.833 V peak-peak
d	-6.833 V peak
Ans	d
15	An inverting amplifier using an Op Amp with R_1 and R_f as the resistors provide an output voltage equal to one of them as
a	$V_0 = -\left(1 + \frac{R_f}{R_1}\right) V_{in}$
b	$V_0 = \left(1 + \frac{R_f}{R_1}\right) V_{in}$
c	$V_0 = -\left(\frac{R_f}{R_1}\right) V_{in}$

d	$V_0 = \left(\frac{R_f}{R_1}\right) V_{in}$
Ans	c
16	A sine wave of 0.5 V peak voltage is applied as an input to a non- inverting amplifier with R_1 = 6 kΩ and R_f = 24 kΩ The output voltage Vo is
a	2.5 V peak
b	-2.5 V peak
c	10 V peak
d	2.5 V peak-peak
Ans	a
17	An ideal OP-AMP has following characteristics
a	R_i = infinity, A = infinity, R_o = 0
b	R_i = 0, A=infinity, R_o = 0
c	R_i= infinity, A = 0, R_o = infinity
d	R_i= 0, A = infinity, R_o = infinity
Ans	a
18	A non-inverting amplifier using Op Amp with R_1 and R_f as the resistors provides an output voltage equal to
a	$V_0 = -\left(\frac{R_f}{R_1}\right) V_{in}$
b	$V_0 = -\left(1 + \frac{R_f}{R_1}\right) V_{in}$
c	$V_0 = \left(\frac{R_f}{R_1}\right) V_{in}$
d	$V_0 = \left(1 + \frac{R_f}{R_1}\right) V_{in}$
Ans	d
19	A non-inverting Op-Amp has a gain of 91 with R_1 = 1 kΩ. The resistor in the feedback path must have a value of
a	91 kΩ
b	90 kΩ
c	99 kΩ
d	92 kΩ
Ans	b
20	For an Op-Amp having differential gain A_d and Common mode gain Ac, CMRR is
a	A_d + Ac
b	1+ (A_d/Ac)
c	A_d/Ac
d	Ac/A_d
Ans	c
21	A non-inverting Op-Amp has a gain of 61 with R_1 = 2 kΩ. The resistor in the feedback path must have a value of
a	61 kΩ

b	62 kΩ
c	121 kΩ
d	120 kΩ
Ans	d
22	Op Amp Comparator is a circuit whose output voltage switches between
a	$+V_{in}, -V_{in}$
b	$+V_o, -V_o$
c	$+V_{sat}, -V_{sat}$
d	None
Ans	c
23	An inverting Op-Amp has a gain of -61 with R_1 = 1 kΩ. The resistor in the feedback path must have a value of
a	61 kΩ
b	60 kΩ
c	59 kΩ
d	62 kΩ
Ans	a
24	With reference to the output voltage of an inverting Op Amp summer $V_0 = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$, if $R_f = R_1 = R_2 = R_3 = R$, then
a	$V_0 = -(V_1 + V_2 + V_3)$
b	$V_0 = -V_1 + V_2 + V_3$
c	$V_0 = V_1 + V_2 + V_3$
d	None of these
Ans	a

UNIT -IV

1	In Colpitts' oscillator, the components used in the feedback network are
a	2R and 2C
b	2L and 1C
c	2C and 1L
d	2L and 2C
Ans	c
2	With a resistance value of R=1kΩ in a feedback network of RC oscillator, frequency of oscillations generated is 5 kHz. The value of the capacitor C is
a	0.0129 μF
b	0.129 μF
c	0.0219 μF
d	129 μF
Ans	a
3	Gain with negative feedback is given by A_f=A/(1+Aβ). The feedback factor is

a	A
b	β
c	A_f
d	Aβ
Ans	b
4	An amplifier has an open loop voltage gain of 1000. If 10% negative voltage series feedback is used, then the closed loop gain is
a	9.9
b	99.9
c	0.9
d	990
Ans	a
5	For an amplifier with negative feedback, the closed loop gain is given by
a	$A_f = \frac{A}{1 - A\beta}$
b	$A_f = \frac{A}{1+A\beta}$
c	$A_f = \frac{A}{1-\beta}$
d	$A_f = \frac{A}{1 - V\beta}$
Ans	b
6	In a practical oscillator circuit, to start oscillations, the loop gain Aβ must be
a	Not Equal to 1
b	Equal to 1
c	Less than 1
d	Greater than 1
Ans	a
7	An amplifier has an open loop voltage gain of 2000. If 40% negative voltage series feedback is used, then the closed loop gain is
a	249
b	24.9
c	0.249
d	2.49
Ans	d
8	In an amplifier, positive feedback leads to
a	Noise
b	Amplification
c	Breakdown
d	Oscillations
Ans	d
9	For an amplifier with positive feedback, the closed loop gain is given by

a	$A_f = \frac{A}{1-\beta}$
b	$A_f = \frac{A}{1+A\beta}$
c	$A_f = \frac{A}{1-A\beta}$
d	$A_f = \frac{A}{1-V\beta}$
Ans	c
10	What is an oscillator?
a	A rectifier
b	A generator
c	An amplifier with positive feedback
d	An amplifier with negative feedback
Ans	c
11	An amplifier has an open loop voltage gain of 100,000. If the negative voltage series feedback factor is 0.01, then the closed loop gain is
a	9.9
b	99.9
c	0.9
d	990
Ans	b
12	In an amplifier with positive feedback, open loop gain A is 20 with feedback factor β as 0.04. Then the gain of the amplifier with feedback is
a	200
b	50
c	100
d	infinity
Ans	c
13	The feedback factor of Colpitt's oscillator is given by
a	$\beta = \frac{A_f}{A}$
b	$\beta = \frac{C_1}{C_2}$
c	$\beta = \frac{A}{A_f}$
d	$\beta = \frac{C_2}{C_1}$
Ans	d
14	Which of the following is not an advantage of a negative feedback amplifier?
a	Higher input impedance
b	Unstable gain
c	Reduction in noise
d	Lower output impedance

Ans	b
15	An amplifier has an open loop voltage gain of 10,000. If the negative voltage series feedback factor is 0.01, then the closed loop gain is
a	990
b	9.9
c	0.99
d	99
Ans	d
16	The feedback factor of Hartley oscillator is given by
a	$\beta = \frac{A}{A_f}$
b	$\beta = \frac{L_2}{L_1}$
c	$\beta = \frac{L_1}{L_2}$
d	$\beta = \frac{A_f}{A}$
Ans	c
17	Which among the following parameters acts as an initiator for the operation of an oscillator in the absence of input signal?
a	Noise voltage
b	Noise power
c	Noise current
d	Noise temperature
Ans	a
18	In which type of oscillator circuit capacitor split representation can be seen in tank circuit?
a	Wein bridge
b	RC phase shift
c	Hartley
d	Colpitts
Ans	d
19	In an RC phase shift oscillator, the resistances in the feedback network are 4.7 kΩ and capacitor values are C = 0.47μF. The frequency of oscillations is
a	0.29413 Hz
b	294.13 Hz
c	2941.3 Hz
d	29.413 Hz
Ans	d
20	The tank circuit of a Colpitt's oscillator has L= 5 mH with C1 = 22.22 nF and C2 = 2.22 nF. The feedback factor β is
a	0.001
b	0.1
c	0.01

d	1
Ans	b
21	An RC network in an RC phase shift oscillator has C = 0.1 μF. The frequency of oscillations is 1 kHz. The value of R used in the feedback network is
a	650 Ω
b	500 Ω
c	600 Ω
d	1 KΩ
Ans	a
22	RC phase shift oscillator is a
a	Low frequency oscillator
b	High frequency oscillator
c	Stable frequency oscillator
d	Relaxation oscillator
Ans	a
23	In a Hartley oscillator, if L1 = 5 mH, L2 = 10 mH and C = 0.01μF, the value of the feedback factor β is
a	50
b	0.05
c	0.5
d	500
Ans	c
24	The frequency of oscillations generated by RC phase shift oscillator is
a	$\frac{1}{2\pi RC}$
b	$\frac{1}{2\pi\sqrt{6}RC}$
c	$\frac{1}{2\pi\sqrt{6RC}}$
d	$\frac{1}{2\pi 6RC}$
Ans	b
25	An RC network in an RC phase shift oscillator has C = 0.2 μF. The frequency of oscillations is 1 kHz. The value of R used in the feedback network is
a	325 Ω
b	500 Ω
c	600 Ω
d	1KΩ
Ans	a
26	Hartley oscillator is a
a	Relaxation oscillator
b	Low frequency oscillator
c	Stable frequency oscillator

d	High frequency oscillator
Ans	d
27	In a Hartley oscillator, if L1 = 7 mH, L2 = 10 mH and C = 0.01μF, the value of the feedback factor β is
a	0.7
b	0.07
c	70
d	700
Ans	a
28	The frequency of oscillations generated by Colpitts oscillator is
a	$\frac{1}{2\pi LC}$
b	$\frac{1}{2\pi\sqrt{L_{eq}C}}$
c	$\frac{1}{2\pi\sqrt{L}}$
d	$\frac{1}{2\pi\sqrt{LC_{eq}}}$
Ans	d
29	The components in the feedback network of Hartley and Colpitt's oscillators are
a	L & C components
b	R & C components
c	Only L Component
d	None of these
Ans	a
30	What is an angle of phase shift for each designed RC network in the Phase Shift Oscillator circuit?
a	30^0
b	45^0
c	60^0
d	90^0
Ans	c
31	In a Hartley oscillator, if L1 = 2 mH, L2 = 8 mH and C = 0.01μF, the value of the feedback factor β is
a	25
b	0.025
c	250
d	0.25
Ans	d
32	The frequency of oscillations generated by Hartley oscillator is
a	$\frac{1}{2\pi\sqrt{C}}$

b	$\frac{1}{2\pi\sqrt{LC_{eq}}}$
c	$\frac{1}{2\pi\sqrt{L_{eq}C}}$
d	$\frac{1}{2\pi LC}$
Ans	c
33	In an RC-Phase shift oscillator, the components used in the feedback network are
a	4R and 4C
b	1R and 1C
c	3R and 3C
d	2R and 2L
Ans	c
34	In a Colpitts oscillator, if C1 = 100 pF, C2 = 60 pF and L = 0.422 H, the value of the feedback factor β is
a	0.6
b	0.06
c	60
d	600
Ans	a
35	The criterion that determines the mathematical condition to generate sustained oscillations is
a	Pinch off
b	Shockley
c	Barkhausen
d	Threshold
Ans	c

UNIT -V

1	Fiber optic cable is a ---------- type of channel.
a	Wireless channel
b	Free space channel
c	Wired channel
d	Radio channel
Ans	c
2	The mechanism of using the same frequency band within a geographical area in a Cellular or mobile communication system is referred to as
a	Frequency reuse
b	Efficiency
c	Reliability
d	Bandwidth coordinator
Ans	a

3	Which of the following is true about microprocessors?
a	It has interfacing circuits
b	It has an internal memory
c	It contains ALU, CU, and registers
d	It uses Harvard architecture
Ans	c
4	Best example for natural noise is
a	Rain
b	Traffic noise
c	Industrial noise
d	Sound pollution
Ans	a
5	The radio channel used for transmission of information from a base station to a mobile station is called,
a	Reverse channel
b	Control channel
c	Mobile channel
d	Forward channel
Ans	d
6	Which one of the following offers external chips for memory and peripheral interface circuits?
a	Microcontroller
b	Embedded system
c	Peripheral System
d	Microprocessor
Ans	d
7	Best example for man-made noise is
a	Rain
b	Radiations from the sun, stars
c	Traffic noise
d	Reflections from Earth
Ans	c
8	In a cellular radio system, the _____ connects the cellular base stations and mobile stations to the Public Switched Telephone Networks.
a	Base station(BS)
b	Mobile Switching Center (MSO)
c	Forward Control Channel (FCC)
d	Time Division Multiple Access (TDMA)
Ans	b

9	The audio frequency range is
a	30 to 34 kH
b	300 Hz to 3.4 kHz
c	20 Hz to 20 kHz
d	10 to 10 kHz
Ans	c
10	The speech signal frequency range is
a	20 Hz to 20 kHz
b	30 to 34 kHz
c	10 to 10 kHz
d	300 Hz to 3.4 kHz
Ans	d
11	Which of the following is true about an Embedded System?
a	An embedded system usually performs a specialized operation and does the same repeatedly
b	It must have a memory, as its software usually embeds in ROM
c	It must have connected peripherals to connect input and output devices.
d	All of them
Ans	d
12	The following frequency band is utilized in cellular or mobile communication system,
a	Audible frequency
b	Audio frequency
c	Unlicensed frequency
d	Radio frequency
Ans	d
13	The loss in the transmission is minimum for the following type of wire
a	Flat cable
b	Copper wire
c	Fiber optic cable
d	Coaxial cable
Ans	c
14	On which component is modulation performed?
a	Transmitter
b	Signal
c	Jammer
d	None of the above
Ans	a
15	What is modulated signal?

a	It is the product of modulating signal and carrier signal
b	It is the product of modulating signal and information signal
c	It is an input signal
d	None of the above
Ans	a
16	What is the primary difference between a microcontroller and a microprocessor?
a	Microcontrollers have higher clock speeds than microprocessors.
b	Microcontrollers have built-in memory and peripherals.
c	Microprocessors have built-in ADCs and DACs
d	Microprocessors are smaller in size
Ans	b
17	Which channel type is used for initiating mobile calls from the mobile device to base station?
a	Forward Control Channel (FCC)
b	Forward Voice Channel (FVC)
c	Reverse Control Channel (RCC)
d	Reverse Voice Channel (RVC)
Ans	c
18	The soft handover of an active call to a new frequency band when a mobile user crosses a cell boundary.
a	Hand off
b	Channel efficiency
c	Frequency reuse
d	Frequency Division Duplexing
Ans	a
19	What is the primary reason mobile communication is referred to as 'Cellular communication'?
a	It uses cells to store energy for mobile devices
b	The communication signals are powered by biological cells
c	It involves mobile devices and a base station within a small geographical area called a cell.
d	It is designed to mimic the structure of biological cells.
Ans	c
20	In a cellular system, a 'cluster' is defined as:
a	A single base station serving multiple mobile users.
b	A group of cells that collectively use the complete set of available frequencies.
c	The central Mobile Switching Centre (MSC) and all its connected base stations.
d	The transmission media connecting the transmitter and receiver.
Ans	b
21	What is the primary purpose of ROM in an embedded system?

a	To store the code or program that the user cannot change
b	To store temporary data for processing
c	To allow users to change the firmware frequently
d	To interface with external communication modules
Ans	a
22	What is the function of a Sensor in an embedded system?
a	To convert signals into physical action
b	To display information to the user
c	To process data without interacting with the real world
d	To convert energy from one form to another for measurement or control, acting as an input device
Ans	d
23	Which of the following is an example of an Actuator?
a	Electric motor
b	Humidity sensor
c	PIR (passive infrared) sensor
d	Keypad
Ans	a
24	The I/O subsystem of an embedded system facilitates interaction with the external world primarily through:
a	Only memory modules
b	Only the core processor
c	Communication interfaces like Wi-Fi or Bluetooth.
d	Sensors and actuators connected to the input and output ports
Ans	d